

helix-code-1.2.0-dev-0.0.1

HelixCode Release Changelog — helix-code-1.2.0-dev-0.0.1

Revision: 2 **Last modified:** 2026-07-28T15:21:59Z **Maintainer:** HelixCode release engineering (documentation draft, AI-agent authored; independently reviewed per §11.4.142/§11.4.194 — see “Independent review” section below)

DRAFT — pinned to a stated HEAD, this is not a cut tag. No git tag has been created by this document. Everything below was derived by reading real git history (`git log`, `git show`) on 2026-07-28 at approximately 13:11 UTC. Multiple background work streams were actively landing commits in this repo and its submodules while this draft was written (constitution §11.4.103 / §11.4.192 continuous multi-track work), so HEAD will have moved by the time this is read. **Re-derive the commit ranges below before treating any of this as final** — do not copy the SHAs or counts forward without checking.

Repo	Stated HEAD (this draft)	Relative to its configured remotes
main (this repo)	c29e1dcc9fad029f64a8836148d8d5ddfc32f0fb	8 commits ahead, 0 behind — not pushed
submodules/ helix_agent	69a5ab8dc2469f73ddf3afa0bced01103df8b656	2 commits ahead of its remotes; working tree dirty (14 modified, 4 untracked)
	8cec90baadad281eb26a5671b792921f03eb7386	

Repo	Stated HEAD (this draft)	Relative to its configured remotes
submodules/ tool_schema		1 commit ahead of its remotes; working tree clean

Format note: docs/changelogs/ did not exist in this repository before this draft — it was created now. The repository’s existing root CHANGELOG.md uses a ## [tag] – date per-release section with Added / Changed / Fixed / Known-gaps groups; this draft follows that convention but adds explicit short-SHA citations per entry and a per-repo commit inventory, so every claim traces back to a real, readable commit (constitution §11.4.6 — no invented history). Per the task scope, nothing here was committed with `git add -A` or similar — only this one file is staged, by explicit path.

Version

helix-code-1.2.0-dev-0.0.1 — draft, not yet tagged.

Prefix resolves to helix-code from `.env HELIX_RELEASE_PREFIX` (§11.4.151).

Prior tags in this lineage:

Tag	Date
helix-code-1.0.0-dev-0.0.1	2026-07-08
helix-code-1.1.0-dev-0.0.1	2026-07-09
helix-code-1.1.0-dev-0.0.2	2026-07-13
helix-code-1.1.0-dev-0.0.3	2026-07-24

On the version jump: helix-code-1.2.0-dev-0.0.1 is a **1.1.x → 1.2.0 minor bump**, not the next -1.1.0-dev-0.0.4 patch in the existing sequence. This was named explicitly by the operator as this release’s target tag; per instruction for this task, that is recorded here as an **operator decision**, not re-litigated or treated as unexplained drift.

Summary — what this release actually delivers for a user

- **Model identity is now honest end-to-end.** A client asking for an alias (e.g. `"model": "default"`) previously got that same alias echoed back in the response, on both the OpenAI-compatible and Anthropic-compatible wires, streaming and non-streaming. The response now names the model that actually served the request (e.g. the real `.gguf` path), matching `CONST-036/037`. Live SSE evidence was captured against a real running coder backend for both wires. One honest exception remains: Anthropic's streaming `message_start` event fires before any chunk exists, so it still carries the requested model — documented in-source, not silently left as a gap.
- **tool_schema translations now work regardless of where you run from.** Previously, calling into this library's `i18n` from any working directory other than its own module root silently returned raw translation keys instead of translated text (e.g. `toolschema_git_desc_commit` instead of “Create git commit”). Bundles are now compiled into the binary (`go:embed`), so this can no longer happen.
- **A large batch of false test failures in helix_agent is fixed.** A liveness check used to accept *any* HTTP response under 500 as “the server is up” — so when a different local service (LLMsVerifier) held the same default port, its 404 response was read as a healthy HelixAgent, and roughly a dozen test packages failed against the wrong service with nothing actually wrong in the product.
- **The whole platform can now be installed and booted as systemd units** (`./setup.sh`) — HelixCode server, HelixAgent, HelixLLM gateway + coder, infra (Postgres/Redis/Weaviate/Qdrant/ChromaDB/Cognee/Ollama), and now LLMsVerifier — with real HTTP health checks and a verified stop/start cycle, rather than requiring manual, host-specific setup.
- **LLMsVerifier is wired to real provider data and publishing real scores** instead of an empty `{}` — the gateway now routes on measured scores instead of always falling back to its static table.
- **go test ./... no longer breaks on generated challenge output** — a gitignored-but-still-scanned results directory could fail three otherwise unrelated packages depending on what an LLM happened to answer during a challenge run.
- **helix_agent boot no longer requires ChromaDB/Cognee on their compiled-in ports**, and no longer hangs indefinitely trying to start a vector store that is already running under a different owner (committed in `helix_agent` but see the pinning note below).

Fixed

SHA	Repo	Summary
e330149f	main	LLM response now reports the model that actually served the request (OpenAI + Anthropic wires, streaming + non-streaming), not the requested alias. Four layers: <code>llm.LLMResponse.Model</code> field + JSON round-trip, <code>OpenAICompatibleProvider</code> populating it from both non-stream and per-chunk stream responses, and both wire facades preferring the served model. Guards at <code>internal/server/wire_facade_model_identity_test.go</code> and <code>internal/llm/openai_compatible_stream_model_test.go</code> , each proven to fail when its fix is reverted.
66d6fb29	main	<code>go test ./...</code> no longer descends into <code>tests/e2e/challenges/test-results/</code> (renamed to <code>_test-results/</code> , the Go toolchain's own "not a package" convention) and tries to compile an LLM's prose answer as Go source. Unblocked 3 packages.
3268fd73	helix_agent (pinned)	<code>testutil.ServerAvailable</code> now requires a real <code>HelixAgent</code> -shaped health payload (HTTP 200 + identity check), not just "any status under 500". Fixes a false-failure cluster of roughly a dozen packages caused by a different local service (<code>LLMsVerifier</code>) answering on the same default port. Three additional call sites that bypassed the shared helper entirely (a raw TCP dial, an error-only skip check) were also fixed.
0165ab1d	helix_agent (pinned)	Precondition gate's container-ownership check fixed in both directions: it no longer flags an unrelated third-party Postgres as "our" duplicate instance (false positive from <code>{port, port+10000}</code> guessing), and it no longer silently no-ops because it ran the wrong runtime command (false negative). Ownership is now established from the actual container runtime, filtered to this project's own containers.

SHA	Repo	Summary
0dfa69e1	helix_agent (<i>pinned</i>)	HelixAgent boot no longer hardcodes ChromaDB (:8001) / Cognee (:8000) endpoints — both resolve via HELIXAGENT_DEP_HOST / HELIXAGENT_PORT_CHROMADB / HELIXAGENT_PORT_COGNEE with the old literals as fallback. Vector-store startup now checks reachability first instead of unconditionally trying to become a second owner of an already-running store, which had been blocking the HTTP listener bind indefinitely.
8cec90b	tool_schema (<i>pinned</i>)	Translation bundles are go:embed-ed, so tr() lookups no longer depend on the process's current working directory. Previously any consumer calling in from outside this module's own root got raw key strings back instead of translations.
69a5ab8d	helix_agent (committed, NOT yet pinned — see Known gaps)	Fixes a goroutine leak in the Ollama provider's CompleteStream: 4 of 6 channel-send sites had no ctx.Done() escape hatch, so a caller that cancelled its context and stopped draining left the background goroutine parked forever. RED/GREEN regression guard added, driven via a real stuck-HTTP-response scenario and a live goroutine-stack inspection.
a345c551	helix_agent (committed, NOT yet pinned — see Known gaps)	Two tests/unit test defects fixed (product behavior was correct in both cases): TestCompletionHandler_Models now wires a real model source instead of asserting on an intentionally-empty list; TestCompletionHandler_Stream now distinguishes an honest empty response (client-gone, request-timeout path) from an actual failure, instead of treating the httptest recorder's default 200 as an unconditional pass signal.
dd3c0c3b	main	Closes the third confirmed instance of an “unprioritized select” race this session: autoSaveLoop's per-tick select between <- ticker.C and <- stop had no priority ordering, so a coincidentally-pending tick could still fire one more SaveAll() after

SHA	Repo	Summary
		DisableAutoSave() had already closed stop — the cause of TestAutoSaveLifecycle_ReEnableTicks’s “disabled auto-save must not tick” flake. Fixed by extracting the per-iteration select into autoSaveTick(tickerC, stop) with a non-blocking priority pre-check on stop. Full RED/GREEN/pre-fix/post-fix polarity evidence captured (§11.4.115): 2488/5000 forced-race hits pre-fix with the priority check disabled, 0/5000 post-fix. Found missing from this changelog’s original draft during independent review — added in revision 2; see “Independent review” section below.

Added

SHA	Repo	Summary
c4115c73	main	./setup.sh now installs and boots the entire platform as systemd units (helix.target + 5 services) — HelixCode server, HelixAgent, HelixLLM gateway, HelixLLM coder, and infra — with a documented port map, boot-persistence via Linger=yes, and a verified full stop/start cycle. Fixes a KillMode=control-group bug that had been silently killing all ten infra containers’ port forwarders whenever the infra unit’s own health check failed.
d3e2c6e7	main	

SHA	Repo	Summary
		Adds LLMsVerifier as a 7th systemd unit, wired to the gateway. Implements the <code>/api/scores</code> endpoint upstream in the submodule (project-agnostic; consumer supplies the port via the unit). Gateway's log changed from "verifier unreachable" to actually reaching the verifier and correctly falling back on an empty score map.
98db7d79	main	LLMsVerifier is now seeded with a real, reachable provider (llamacpp → the local llama.cpp coder server) via a consumer-owned <code>config/llmsverifier/config.yaml</code>, instead of the submodule's own empty <code>llms: []</code> default. First real verification score published (0.96) from genuine inference, replacing the previously-empty <code>/api/scores</code>.
410afda5	helix_agent (<i>pinned</i>)	Hyper provider discovery, registry, and verifier types.
1296d328	main	Adds <code>scripts/setup.sh</code> scaffolding (157 lines) and two systemd unit files (<code>helixcode-infra.service</code>, <code>helixllm-</code>

SHA	Repo	Summary
		gateway.service) — precursor work later built out by c4115c73 / d3e2c6e7. Commit message is the literal Auto-commit; see Known gaps.

Changed

SHA	Repo	Summary
925aa859	main	Backfills the constitution's §11.4.182 TRUNK RULE (work on trunk is always labelled Track 1) across this project's five root governance carriers, and bumps the helix_agent and tool_schema submodule pins to the commits described above.
f14a3f89	main	Bumps the helix_agent submodule pointer (Hyper provider).
565506f6	main	Bumps HelixLLM to helixllm-0.2.0 (CPU inference).
c20215bd	helix_agent (<i>pinned</i>)	Chaos/compliance/automation test hardening carried in from prior sessions and committed so it is not lost — chaos/api, chaos/auth, module-compliance, config-generation automation, plus new helix_endpoint

SHA	Repo	Summary
		contract tests. go build ./... verified exit 0; the commit itself is explicit that the full suite was not independently reviewed or proven green at the time.
4b5f2867	helix_agent (<i>pinned</i>)	gofmt alignment and comment-spacing pass across the module.
36597718	helix_agent (<i>pinned</i>)	Search test reconciled to a reordered UnknownType check; adds coverage for the reachability short-circuit.

Infrastructure / Governance / QA evidence

SHA	Repo	Summary
578de68d	main	Lands §11.4.83 end-user QA evidence for 7 feature commits, captured against a freshly rebuilt binary (111 assertions, 0 FAIL, 0 SKIP). Explicitly documents that an earlier evidence pass had validated a 15-day-stale binary and was therefore invalid and re-run.
1c9fdcf2	main	§11.4.83 evidence against live paid providers (Cerebras, Cohere) and a real MCP wire for 3 further commits; drops the qa-evidence-gate violation count from 6 to 3.
e952f4d1	main	§11.4.83 evidence for the last 3 non-feature commits (CORS allowlist, Fyne desktop build

SHA	Repo	Summary
		docs, cognee determinism), each carrying an explicit “honest scope statement” about limited/no end-user surface. Drives the qa-evidence gate (G7) from 13 to 0 violations across 28 feature-shipping commits.
d8c44618	main	\$11.4.83 evidence specifically for the e330149f model-identity fix (8/8 assertions PASS, mutation table proving each guard fails when deleted).
a5462e35	main	Live SSE proof, against the real coder backend, that <code>stream:true</code> reports the served model on every chunk (13 frames captured, 1 distinct model id, 0 frames carrying the alias).
c29e1dcc	main	Live proof for the Anthropic wire (POST /v1/messages returns the real served-model path) and for the <code>tool_schema</code> i18n fix (a compiled consumer binary resolves translations from unrelated working directories).
cf26a173	main	Bumps 4 owned submodule gitlinks (constitution, challenges, helix_qa, helix_agent) after pushing them to their upstreams. Commit itself states the full test sweep was not green at the time and no tag was cut.
0a4eb8d0	main	Corrects stale live-state anchors in <code>RESUME.md</code> after the above push — the exact “stale handoff doc” trap the project’s own governance

SHA	Repo	Summary
		warns about, caught and fixed in the same session.
56f7edf7	main	Message is the literal Auto-commit (no descriptive body). Diffstat shows real, substantive content: nil-request guards for the Replicate and Together LLM providers, a malformed-URL guard for the llamacpp provider, a security_scan tool rewrite, and an overhaul of verify_qa_evidence.sh / verify-governance-cascade.sh / obsolete_details_gate.sh / summary_sync_gate.sh, plus a helix_agent pointer bump. See Known gaps.

Per-repo commit inventory

main (this repo) — 19 commits since helix-code-1.1.0-dev-0.0.3 (5ab97d8c)

(Revision 2: corrected from 18 to 19 — dd3c0c3b was omitted from the original draft despite being this changelog’s own direct parent commit. See “Independent review” below.)

```

dd3c0c3b fix(persistence): close unprioritized-select race in autoSaveLoop
c29e1dcc docs(qa): LIVE proof for the Anthropic wire and the i18n cwd fix
a5462e35 docs(qa): LIVE proof that stream:true reports the served model
e952f4d1 docs(qa): §11.4.83 evidence for the three non-feature commits – G
1c9fdcf2 docs(qa): §11.4.83 evidence for the Cerebras, Cohere and MCP-conc
d8c44618 docs(qa): §11.4.83 evidence for e330149f model-identity fix
925aa859 docs(governance): backfill the §11.4.182 TRUNK RULE + bump two su
e330149f fix(llm/server): report the model that ACTUALLY served the reques
578de68d docs(qa): land §11.4.83 end-user QA evidence for 7 feature commit
0a4eb8d0 docs(resume): correct live-state anchors after the end-of-session
cf26a173 chore(submodules): bump gitlinks after end-of-session sync
56f7edf7 Auto-commit
66d6fb29 fix(e2e): stop go test compiling generated challenge output (3 pa
98db7d79 feat(llmsverifier): seed real providers via consumer-owned config

```

d3e2c6e7 feat(systemd): add LLMsVerifier as the seventh unit, wired to the
c4115c73 feat(systemd): install and boot the whole platform – HelixCode +
1296d328 Auto-commit
565506f6 chore: bump HelixLLM to helixllm-0.2.0 (CPU inference)
f14a3f89 chore: bump HelixAgent (Hyper provider)

submodules/helix_agent — 9 commits since the previously-pinned 55ad06c6

Pinned into main's current gitlink (3268fd73) — 7 commits:

3268fd73 fix(testutil/tests): verify the server is HelixAgent, not just th
0165ab1d fix(precondition): establish container ownership from the runtime
c20215bd test(helix_agent): chaos/compliance/automation hardening + live-e
4b5f2867 style: gofmt alignment and comment spacing across helix_agent
36597718 test(search): reconcile UnknownType assertion to the reordered ty
0dfa69e1 fix(boot): resolve dependency endpoints from env; don't block sta
410afda5 feat: Hyper provider discovery + registry + verifier types

Committed in helix_agent's own history but NOT yet reflected in main's gitlink — 2 commits (see Known gaps):

69a5ab8d fix(ollama): close goroutine-leak in CompleteStream on ctx cancel
a345c551 fix(tests/unit): wire the model source; stop reading recorder-def

submodules/tool_schema — 1 commit since the previously-pinned cb7bf675, fully pinned

8cec90b fix(i18n): embed translation bundles so lookups work from any dire

Known gaps / not yet verified

This section exists so this document does not read as more finished than the underlying work is. None of these were papered over.

1. **§11.4.185 manual QA-team confirmation has not been given.** No record of a manual QA sign-off for this scope exists anywhere in this repository. Per this project's own governance, automated green is necessary but not sufficient — a manual QA pass is required before this scope can be called fully complete, and it has not happened.

2. **Desktop GUI packages do not build on this host.** Directly attempted `go build ./applications/...` in `helix_code/` on this host and it fails with two real, host-environment errors, not code defects:
 - `github.com/go-gl/gl/v2.1/gl: pkg-config cannot find gl.pc.`
 - `github.com/go-gl/glfw/v3.3/glfw: fatal error: X11/Xlib.h: No such file or directory.` This is not a new discovery — a prior commit in this cycle (`d6c05f76`, already merged before this release's commit range) documented this same gap and verified the README's claimed error text is byte-identical to the real one on this host. Re-confirmed here, not re-litigated. Any package depending on the Fyne desktop GUI stack is affected.
3. **The zero-skip full integration test stack was not booted for the last sweep.** Checked directly (`podman ps`): the containers currently running on this host (`helixcode-infra-postgres`, `-redis`, `-weaviate`, `-qdrant`, `-chromadb`, `-cognee`, `-ollama`, etc., all `systemd`-managed via `compose.helixcode-infra.yml`) are the **production-like platform infra**, not the `docker-compose.full-test.yml` zero-skip test stack (which would run containers named `helixcode-postgres-full`, `helixcode-redis-full`, etc. — none of those are present). Any integration-test result gathered without first running `make test-infra-up` for that specific stack is environmental, not a confirmed pass or fail, until it is booted and the sweep re-run against it.
4. **The full go test suite has not concluded cleanly on a quiet host for this cycle.** `docs/CONTINUATION.md`'s own rev-18 audit (not independently re-run as part of this documentation-only task, to avoid adding more load to a host other agents are actively building/testing on) records one completed `helix_agent` full-suite run on a contended host reporting 18 FAIL packages, and attributes this to ambient host contention (constitution §11.4.119) consistent with this project's own documented pattern — but that specific 18-package list has not itself been re-run on a quiet host and confirmed clean. Treat as **unconfirmed**, not cleared.
5. **`helix_agent`'s goroutine-leak fix (69a5ab8d) and test fix (a345c551) are real, committed work but are NOT yet part of this release's pinned artifact.** Main's current gitlink for submodules/`helix_agent` points at `3268fd73`; the submodule's own HEAD is two commits further, at `69a5ab8d`. Until the gitlink is bumped and re-verified, this release does not actually ship the Ollama goroutine-leak fix even though the fix exists and is committed.
6. **`helix_agent`'s working tree is dirty** (modified tracked files and untracked new test files observed at read time) and **2 commits ahead of its own**

configured remotes — i.e. not yet pushed anywhere outside this host. This draft does not push anything.

7. **main is 8 commits ahead of every configured remote** (0 behind), i.e. also not yet pushed. This document commits only itself; per this task's constraints, nothing is pushed and no tag is created.
8. **Two commits carry the literal message Auto-commit** (56f7edf7, 1296d328) with no descriptive body. Real, substantive content was recovered from their diffstats (see the Infrastructure / Added tables above) rather than invented, but the commit messages themselves do not describe intent, verification, or evidence — a documentation-clarity gap at the commit-message layer, flagged rather than silently backfilled.
9. **No claim in this document was accepted from a stale project artifact without independent re-derivation of the specific git facts it depends on** (HEAD SHAs, gitlink pins, tag list, container names) — those were each re-checked directly against the live repositories at draft time. The one exception is item 4 above, which is explicitly attributed to docs/CONTINUATION.md rather than re-run, and is labeled unconfirmed for exactly that reason.
10. **helix_agent has moved further since this draft's stated pin (69a5ab8d), adding a third sibling to the same defect class this document already tracks.** Commit e2f42b12 (fix(llm,database): prioritize cancellation in racy select statements) fixes exactly the two “sibling” locations dd3c0c3b's commit message names as instances of the same unprioritized-select pattern — internal/llm/lazy_provider.go's createProviderWithContext and internal/database/debate_log_repository.go's StartCleanupWorker — with the same RED/GREEN polarity-evidence discipline (§11.4.115). It is real, committed work, but as of this revision it is **not yet part of this release's pinned artifact**: main's gitlink for submodules/helix_agent still points at 3268fd73, four commits behind helix_agent's own current HEAD (3268fd73 → a345c551 → 69a5ab8d → e2f42b12). This compounds Known gap 5 rather than replacing it — do not treat 69a5ab8d as the current unpinned tip; re-derive submodules/helix_agent's actual HEAD before citing it, per this document's own opening warning.

Independent review

Per constitution §11.4.142 (every change gets an independent review, no exception) and §11.4.194 (exhaustive all-scenario review), this document received a full independent review after its original commit (8bdac5b0) landed. The authoring agent had been killed by an API session limit immediately after committing and never reported, so this changelog had zero review until now.

Method: every cited short SHA in the Fixed / Added / Changed / Infrastructure tables and the per-repo commit inventories was checked to exist and to actually do what its entry claims, via `git show --stat` and full `git show` against this repository, submodules/helix_agent, and submodules/tool_schema — not accepted on the strength of the subject line alone. Quantitative claims (assertion counts, PASS/FAIL tallies, live scores, frame counts) were spot-checked against the cited commits’ own bodies. `git log --oneline 8bdac5b0..HEAD` was run in this repo (empty — main has not moved) and, using the changelog’s own stated submodule pins as the base, in submodules/helix_agent (one further commit, e2f42b12 — see Known gap 10) and submodules/tool_schema (none). The `.env` `HELIX_RELEASE_PREFIX` resolution, the §11.4.44 revision header, and the “3 desktop packages fail to build” claim (applications/desktop, applications/aurora_os, applications/harmony_os — confirmed all three import the Fyne stack, so all three are genuinely covered by `go build ./applications/...` and by the stated X11/OpenGL failure) were all independently re-verified rather than taken on trust.

Finding (blocking, now fixed in revision 2): commit dd3c0c3b — this changelog’s own direct parent commit, committed roughly 90 seconds before 8bdac5b0 — was completely absent from every section of the original draft despite being real, substantive, in-scope work covered by the document’s own stated data-collection window. It has been added to the Fixed table, the per-repo commit inventory, and cross-referenced against its helix_agent sibling fix (Known gap 10).

Verdict: no invented history was found anywhere in this document — every other cited SHA, quantitative claim, and Known-gaps disclosure checked out exactly as stated. With the one finding above corrected, this document is independently confirmed accurate as of this revision (2026-07-28T15:21:59Z).

Sources verified

All SHAs, commit subjects, commit bodies, tag names/dates, gitlink pins, container names, and the `go build` error text above were read directly from this repository, submodules/helix_agent, and submodules/tool_schema via `git log`, `git show`, `git tag`, `git ls-tree`, `podman ps`, and a direct `go build` invocation on 2026-07-28. No claim in the Fixed / Added / Changed tables above is unsourced from a real commit; anything that could not be traced to a commit was left out rather than inferred.